

Predicting Trust and Distrust in Bitcoin Web-of-Trust Networks

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Group 3

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Abstract

Predicting the sign of edges in signed directed graphs is a fundamental challenge in network analysis, with direct applications to **fraud detection and risk assessment in decentralised financial systems**.

... Results ...

1 Introduction

Project Overview In this work, we study trust and distrust prediction on two **Bitcoin web-of-trust networks**, Bitcoin OTC and Bitcoin Alpha, framing the problem as binary edge-sign classification on severely class-imbalanced graphs. We propose a feature-engineering pipeline grounded in **Balance Theory, Status Theory, and reciprocity**, and train a **Random Forest classifier** on temporally split data to avoid leakage. We then compare this approach against a **Signed Graph Convolutional Network (SGCN)**, which learns node representations through separate aggregation channels for positive and negative neighbourhoods. An **ablation study** quantifies the marginal contribution of each theoretical feature group, while a **cross-dataset evaluation** tests whether learned trust dynamics generalise across platforms. Models are assessed via **AUC-ROC** and **negative-class F1-score**, metrics chosen for robustness to class imbalance.

Problem Statement Decentralized peer-to-peer platforms such as Bitcoin rely on *web-of-trust* mechanisms in which users assign explicit positive or negative ratings to counterparties. These ratings form a *signed directed graph* whose negative edges carry critical information about potential fraud, adversarial behavior, and systemic risk. Most graph-based models either ignore edge polarity or treat all edges as structurally equivalent. We focus specifically on **sign prediction**: given that an interaction occurs between users u and v , can we predict whether it will be a trust (+) or distrust (−) relationship? Accurate solutions could support early-warning systems for harmful actors in decentralised financial networks.

2 Related work

Sign prediction in signed networks has attracted growing interest as a natural extension of classical link prediction. Early approaches framed the problem through the lens of **social balance and status theories**. Leskovec et al. [1] introduced these two complementary frameworks to explain the formation of positive and negative edges in online com-

munities, showing that **structural patterns derived from triads are strong predictors of edge sign**. Their work on Epinions and Slashdot established the empirical foundation that subsequent models have built upon. Feature-based machine learning approaches have since formalised these intuitions. Wu et al. [2] **applied Random Forest to edge sign prediction on the Wikipedia signed network**, assembling hand-crafted features from degree statistics and triad counts. Their results demonstrated consistent improvements over logistic regression baselines, **validating the utility of sociological features in a supervised classification setting**. Although their domain differs from cryptocurrency networks, the problem structure, binary classification under severe class imbalance, transfers directly to our setting.

Graph neural network methods have more recently superseded purely feature-based approaches. Derr et al. [3] proposed the **Signed Graph Convolutional Network (SGCN)**, **which extends spectral convolution to signed graphs by maintaining separate aggregation channels for positive and negative neighbourhoods**. This architecture encodes balance-theoretic assumptions directly into the learned representations, and has become the **standard deep learning baseline for sign prediction tasks**. While existing work evaluates models across multiple signed networks, each model is typically trained and tested within the same platform using random splits. Cross-dataset transfer, i.e. training on one network and evaluating directly on another without retraining, remains largely unexplored, particularly in the context of cryptocurrency trust networks.

3 Methods

3.1 Data Description

We use two publicly available signed directed graphs from the Stanford SNAP repository [5]: **Bitcoin OTC and Bitcoin Alpha**. Both networks model *web-of-trust* dynamics among anonymous users of Bitcoin trading platforms, where the absence of verifiable identity creates a concrete need for reputation mechanisms. Each directed edge ($u \rightarrow v$) represents an explicit rating assigned by user u to user v , with

integer weights in $[-10, +10]$ and an associated Unix timestamp. For the purpose of sign prediction, we binarise the edge label directly from the **weight**: edges with weight > 0 are labelled as trust (+), and edges with weight ≤ 0 as distrust (−).

Table 1: Summary statistics of the two Bitcoin web-of-trust datasets.

Dataset	Nodes	Edges	Density	Avg Deg.	Avg Clust.
Bitcoin OTC	5,881	35,592	0.001029	6.05	0.1511
Bitcoin Alpha	3,783	24,186	0.001690	6.39	0.1583

Both datasets are extremely sparse, with edge densities below 0.002, which is typical of large-scale web-of-trust networks. Despite this global sparsity, the average clustering coefficients of 0.158 (Alpha) and 0.151 (OTC) indicate that users tend to form locally dense trust communities, consistent with the prevalence of balanced triads observed in the triad census. The near-identical average degrees (6.39 and 6.05 respectively) suggest that the underlying trust dynamics are structurally comparable across the two platforms, supporting the feasibility of cross-dataset generalisation (RQ2).

3.2 Preprocessing & Analysis

Raw edges carry integer ratings in $[-10, +10]$ alongside a Unix timestamp. We derive two quantities from each rating: a binary sign label (+1 if rating > 0 , −1 otherwise) and an absolute weight. From the resulting class distribution it has emerged a **severe imbalance in both networks**: Bitcoin Alpha contains 93.6% positive and 6.4% negative edges, while Bitcoin OTC contains 90.0% positive and 10.0% negative edges.

3.2.1 Temporal Split

It was found that while the *soc-sign-bitcoinotc* dataset was pre-sorted, the *soc-sign-bitcoinalpha* dataset was not. To ensure consistency, we programmatically ordered both datasets by timestamp before partitioning them into training (oldest 80%) and test (most recent 20%) sets. This yielded 19,348 / 4,838 edges for Alpha and 28,473 / 7,119 for OTC respectively. To **ensure scientific validity**, we verify that no test edge precedes the last training edge, preventing any form of temporal leakage. Graphs are then constructed exclusively from training edges using *NetworkX DiGraph* [6], so that **structural features computed at prediction time never incorporate future information**.

3.2.2 Degree Distribution

Figure 1 reports the **log-log degree distributions** for both networks. In-degree and out-degree follow an approximate **power-law distribution** in both datasets, indicating that a small number of highly active raters and highly rated users dominate the network, a characteristic common to real-world web-of-trust systems.

3.2.3 Triad Census

We enumerate all directed triangles in the training graphs following the classification established by [7] and categorize them according to **Balance theory** [8]. A triad is considered

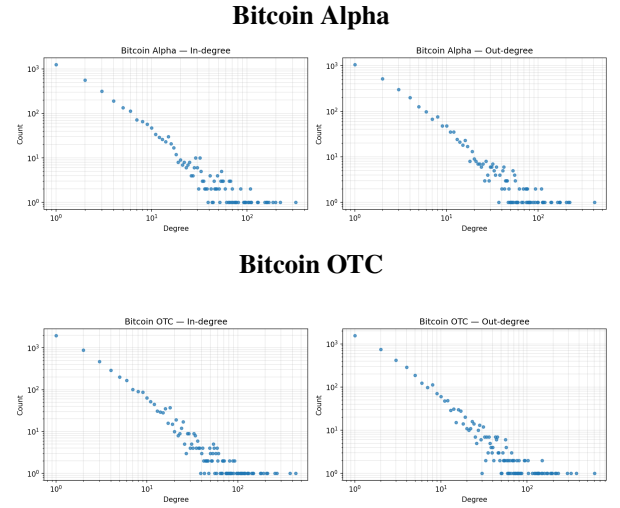


Figure 1: Log-log degree distributions (in- and out-degree) for Bitcoin Alpha and Bitcoin OTC networks.

balanced if the product of its three edge signs is positive (i.e., +++ or +−− configurations). Bitcoin Alpha yields 49,981 triads, of which 91.0% are balanced and 9.0% unbalanced. Bitcoin OTC yields 71,020 triads, with 91.4% balanced and 8.6% unbalanced. The **strong overrepresentation of balanced triads** in both networks provides empirical evidence that **Balance theory actively shapes trust dynamics**, and directly justifies the use of balance-theoretic features in our classifier.

3.3 Feature Engineering

For each candidate edge $(u \rightarrow v, t)$ in both the Bitcoin Alpha and Bitcoin OTC datasets, already ordered chronologically and split 80/20 into training and test sets, a feature vector is constructed to serve as input to the Random Forest classifier. The dependent variable y represents the sign of the interaction: trust (+) or distrust (−). Every feature is computed using only edges with timestamp $< t$, so no future information leaks into the model at any stage. The vector is built from three pillars: (1) local node behaviour, (2) sociological structure, and (3) reciprocity..

3.3.1 Local Node Behaviour

Local node features capture the individual reputation of each endpoint directly from their interaction history, without relying on the global graph structure. For each edge $(u \rightarrow v, t)$, we ask: how has u behaved as a rater in the past, and how has v been perceived by others?

For the source u we extract three features: (1) **out-degree** $d^{\text{out}}(u, t)$, the total number of ratings u has issued before t ; (2) **mean rating given** $\bar{r}^{\text{out}}(u, t)$, the average score u assigns to others; and (3) **positive fraction given** $\rho^{\text{out}}(u, t)$, the share of u 's ratings that are positive. A negative rating from a habitually harsh rater carries less weight than one from a consistently positive user. For the target v we compute: (4) **in-degree** $d^{\text{in}}(v, t)$, the total ratings v has received; (5) **mean rating received** $\bar{r}^{\text{in}}(v, t)$, the average score the community assigns to v ; and (6) **positive fraction received** $\rho^{\text{in}}(v, t)$, where a node receiving for example 90% positive ratings is more trustworthy than one with only 40%.

Nodes with no prior history default to zero for all six features.

3.3.2 Sociological Structure

Balance Theory. Balance Theory [8] predicts that triads tend toward configurations where the product of edge signs is positive. For a candidate edge $(u \rightarrow v, t)$, we identify every common neighbour w such that both $u \rightarrow w$ and $w \rightarrow v$ exist before t . For each such w , the implied sign of $u \rightarrow v$ under balance is $\hat{s} = s_{uw} \cdot s_{wv}$. We summarise this distribution with two features: the fraction of common neighbours that imply a positive sign (`frac.balance.pred.pos`) and a net balance signal `balance.signal` = $(n_+ - n_-) / n_{\text{total}} \in [-1, +1]$. When no common neighbour exists both features default to zero.

Validation confirms the signal is strong: among edges with at least one common neighbour, balance-only sign accuracy reaches 95.0% on Alpha train and 94.9% on OTC train, degrading only slightly to 89.4% and 91.8% on the respective test sets. The fraction of balanced triads is 91.4% (Alpha) and 92.0% (OTC) on train, consistent with the global triad census reported in Section 3.2.3.

Status Theory. Status Theory [1] frames trust as a signal of relative standing: assigning a positive rating implies that the target holds higher status than the source. We define the status of a node x at time t as

$$S(x, t) = d_{\text{in}}^+(x, t) - d_{\text{in}}^-(x, t) \quad (1)$$

where d_{in}^+ and d_{in}^- are the positive and negative in-degrees accumulated before t . The feature for edge $(u \rightarrow v, t)$ is the status differential $\Delta S = S(v, t) - S(u, t)$: a large positive value means v is much more reputable than u , which Status Theory associates with a positive rating.

Validation supports this: on Alpha train, edges labelled positive have a mean ΔS of -1.14 , while edges labelled negative have a mean of -12.08 . The gap widens on the test sets (-1.26 vs -46.86 on Alpha, -4.48 vs -57.57 on OTC), indicating that status separation becomes more pronounced as the network matures.

3.3.3 Reciprocity

Bitcoin Alpha has a reciprocity of 0.85 and Bitcoin OTC of 0.81, indicating that the **large majority of interactions are mutual**. Among reciprocated edges, positive return ratings dominate overwhelmingly: 97.9% in Alpha and 97.0% in OTC. This suggests that **reciprocity is a strong signal of trust consolidation**.

At the edge level, we encode reciprocity with two binary/ordinal features: `has.reverse` $\in \{0, 1\}$, which flags whether v has already rated u before t , and `reverse.sign` $\in \{-1, 0, +1\}$, which records the sign of that prior rating (0 if none exists). Approximately 40% of training edges have a reverse counterpart in both datasets. Among those, positive edges carry a mean reverse sign of ~ 0.43 , while negative edges carry a mean near zero, confirming that mutual trust is strongly asymmetric in its sign distribution.

3.3.4 Final Feature Vector

Table 2: Complete feature set used as input to the Random Forest classifier.

Pillar	Feature	Description
1. Local Node Behaviour	$d^{\text{out}}(u, t)$	Ratings u has issued before t
	$\bar{r}^{\text{out}}(u, t)$	Mean score u assigns to others
	$\rho^{\text{out}}(u, t)$	Fraction of positive ratings u gives
	$d^{\text{in}}(v, t)$	Ratings v has received before t
	$\bar{r}^{\text{in}}(v, t)$	Mean score the community assigns to v
	$\rho^{\text{in}}(v, t)$	Fraction of positive ratings v receives
2. Sociological Structure	n_{common}	Number of common neighbours w with $u \rightarrow w$ and $w \rightarrow v$
	n_+, n_-	Common neighbours implying positive / negative sign
	<code>frac.balance.pred.pos</code>	Fraction of common neighbours implying positive sign
	<code>balance.signal</code>	Net balance vote $(n_+ - n_-) / n_{\text{common}}$
	$\Delta S = S(v, t) - S(u, t)$	Status differential under Status Theory
3. Reciprocity	<code>has.reverse</code>	1 if v has rated u before t , else 0
	<code>reverse.sign</code>	Sign of v 's prior rating of u ; 0 if absent

4 Random Forest Algorithm

Random Forest is a learning method that constructs a collection of decision trees at training time and outputs the class predicted by the majority of trees. Each tree is trained on a bootstrap sample of the training data (sampling with replacement), and at each internal node the **best split is selected** among a random subset of $\sqrt{p}$ features (where p is the total number of features), rather than all available ones. This double source of randomness decorrelates the individual trees and reduces variance without substantially increasing bias.

Formally, given a training set $\mathcal{D} = (\mathbf{x}_i, y_i)_{i=1}^n$ with $y_i \in \{+1, -1\}$, the model builds T trees $h_{t=1}^T$ and predicts:

$$\hat{y} = \text{sign} \left(\sum_{t=1}^T h_t(\mathbf{x}) \right)$$

The input to each tree is the **feature vector** described in ??; the output is a binary edge-sign label (trust or distrust). In our implementation we use `RandomForestClassifier` from the `scikit-learn` library [9].

Algorithm 1 Random Forest Classifier

Require: Training set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, number of trees T , feature vector $\mathbf{x}_i \in R^p$

Ensure: Predicted edge-sign label $\hat{y} \in \{+1, -1\}$

- 1: **for** $t = 1$ **to** T **do**
- 2: Draw a bootstrap sample $\mathcal{D}_t$ from $\mathcal{D}$ with replacement
- 3: At each node, select $\lfloor \sqrt{p} \rfloor$ features at random
- 4: Grow a decision tree h_t by recursively splitting on the most informative feature, until a stopping criterion is met
- 5: **end for**
- 6: **Predict** the sign of a new edge $\mathbf{x}$ by majority vote:

$$\hat{y} = \text{sign} \left(\sum_{t=1}^T h_t(\mathbf{x}) \right)$$

Each tree $h_t : R^p \rightarrow \{+1, -1\}$ takes the feature vector of a candidate edge as input and returns a class prediction. The final label is determined by majority vote across all T trees: if more trees predict trust than distrust the sum is positive and $\hat{y} = +1$, otherwise $\hat{y} = -1$.

5 Remaining Work

The feature engineering pipeline described in Section 3.3 is complete and validated on both datasets. The following steps remain before the final report.

Random Forest classifier. We will train a `RandomForestClassifier` on `X_train` for both Bitcoin OTC and Bitcoin Alpha, using class-weight balancing to address the severe positive/negative imbalance documented in Section 3.2. Hyperparameters will be selected via cross-validation. Performance will be reported as AUC-ROC and negative-class F1-score on the held-out test sets.

Ablation study. To quantify the marginal contribution of each feature group, we will retrain the classifier four times, removing one pillar at a time (local node behaviour, balance theory, status theory, reciprocity). Comparing AUC-ROC across conditions will directly answer RQ1.

Cross-dataset generalisation. To address RQ2, we will train on Bitcoin OTC and evaluate directly on Bitcoin Alpha without retraining. The structural comparability of the two networks, evidenced by their near-identical density and average degree (Table 1), provides a principled basis for this transfer experiment.

Signed Graph Convolutional Network. As a deep learning baseline, we will implement the SGCN architecture [3], which maintains separate aggregation channels for positive and negative neighbourhoods. Results will be compared against the Random Forest on the same temporal splits and evaluation metrics.

Final report. Results from all experiments will be consolidated into a complete 10-page report, including a unified results table, discussion of failure cases, and conclusions on the generalisability of sociological theories to cryptocurrency trust networks.

Use of AI Assistance

During the preparation of this milestone, AI language model tools were used in the following limited capacities: to support understanding of course concepts such as Balance Theory, Status Theory, and graph neural network architectures; to verify the correctness of implementation logic and check for potential sources of data leakage; and to assist in editing draft text for clarity and academic register. All experimental design decisions, mathematical formulations, and interpretations of results are our own. No AI tool was used to generate results, write code autonomously, or produce content presented as original scientific contribution.

References

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